THE SAT FOUNDER'S TRANSFORMATION GUIDE

6-Month Journey: 8 LPA to 30 LPA

Building India's Device Intelligence Platform

For:	Paresh Ranjan Rout
Duration:	180 Days | 360 Hours
Start Date:	January 20, 2025
End Goal:	30 LPA Senior Data Engineer
Startup:	SAT Platform MVP Ready

"I will do whatever hard work required." - Paresh

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CHAPTER 1: YOUR VISION & MISSION

SAT (Secure Device Intelligence) is a nation-impact system design. You are building a lawful, data-driven, India-first platform for device recovery and digital safety.

The SAT Architecture - 8 Layers

Layer 0	Legal Foundation	DPDP Compliance, Permissions
Layer 1	Secure Device Identity	Android ID, Encrypted Binding
Layer 2	Event Collection	SIM Change, Network Shift
Layer 3	Location & Charging	Last Location, Context Only
Layer 4	Reset & Tamper Detection	Factory Reset, Repair Signals
Layer 5	On-Device AI Analysis	Usage Change, AI Alerts
Layer 6	Police & Telecom Intel	FIR Timeline, CEIR Actions
Layer 7	Trust & Safety	Fake FIR Check, Anti-Abuse

"I am a Cloud & Data Platform Engineer. I will master this stack. It will pay me now and power SAT later."

CHAPTER 2: THE 6-MONTH ROADMAP

Month	Focus	SAT Layer	Outcome
1	SQL + Data Thinking	Layer 2	Think like Data Engineer
2	Data Modeling	Layer 1,3	Design schemas
3	Azure Data Factory	Layer 2	Production pipelines
4	PySpark + Databricks	Layer 5	Process millions
5	End-to-End Project	All	SAT MVP
6	Interview Prep	Layer 6,7	30 LPA Ready

Technology Stack (LOCKED):

- SQL (Advanced)
- Azure Data Factory
- Azure Data Lake Gen2
- Databricks (PySpark)
- Azure Synapse
- Python for Data
- GitHub CI/CD

CHAPTER 3: DAILY RULES & PRINCIPLES

The 7 Non-Negotiable Rules

Rule 1: Same Time Every Day

9 PM - 11 PM recommended. Protect this time.

Rule 2: Phone on Airplane Mode

Zero distractions for 2 hours.

Rule 3: 60-40 Split

60 min learning, 60 min hands-on practice.

Rule 4: Daily Notes

Write: What learned, What confused, One real use.

Rule 5: Saturday = SAT Day

Apply weekly learning to SAT project.

Rule 6: Sunday = Review

Review week, plan next week.

Rule 7: Monthly Measurement

Don't measure daily. Review monthly.

"Confusion is not lack of intelligence - it is lack of constraint."

CHAPTER 4: MONTH 1 - SQL MASTERY

Goal: Think like a Data Engineer | SAT: Layer 2 Event Collection

Week 1: SQL Fundamentals (Days 1-7)

Day 1: SELECT, FROM, WHERE basics
Day 2: AND/OR/NOT, IN, BETWEEN operators
Day 3: ORDER BY, LIMIT/OFFSET
Day 4: NULL handling - IS NULL, COALESCE
Day 5: String functions - CONCAT, SUBSTRING
Day 6: Date functions - DATEADD, DATEDIFF
Day 7: ASSIGNMENT: 20 SQL problems

Week 2: JOINs (Days 8-14)

Day 8: INNER JOIN
Day 9: LEFT/RIGHT JOIN
Day 10: FULL OUTER JOIN
Day 11: Self Joins
Day 12: Multi-table joins
Day 13: Join performance
Day 14: ASSIGNMENT: 25 JOIN problems

Week 3: Aggregations (Days 15-21)

Day 15: GROUP BY, COUNT, SUM, AVG
Day 16: HAVING clause
Day 17: Advanced aggregates
Day 18: CASE statements
Day 19: Subqueries intro
Day 20: Correlated subqueries
Day 21: ASSIGNMENT: 25 aggregation problems

Week 4: Advanced SQL (Days 22-30)

Day 22: CTEs - WITH clause
Day 23: Window functions - OVER, PARTITION BY
Day 24: ROW_NUMBER, RANK
Day 25: LAG, LEAD
Day 26: FIRST_VALUE, LAST_VALUE
Day 27: Query optimization
Day 28: ASSIGNMENT: 30 advanced problems
Day 29-30: PROJECT: SAT Event Query System

CHAPTER 5: MONTH 2 - DATA MODELING

Goal: Understand how data should LIVE | SAT: Layer 1 & 3

Week 1: Modeling Fundamentals

- OLTP vs OLAP
- Star Schema
- Dimension Modeling
- Fact Table Design
- Snowflake Schema
- Keys & Relationships

Week 2: Slowly Changing Dimensions

- SCD Type 1, 2, 3
- Time Dimensions
- Junk Dimensions
- Degenerate Dimensions

Week 3: Azure Storage

- File Formats (Parquet)
- Azure Data Lake Gen2
- Medallion Architecture
- Partitioning
- Security

Week 4: Advanced Storage

- Delta Lake
- Data Versioning
- Schema Evolution
- Data Quality
- Cost Optimization

PROJECT: Build SAT Data Warehouse in Azure

CHAPTER 6: MONTH 3 - AZURE DATA FACTORY

Goal: Build production-grade ETL pipelines | SAT: Layer 2 Ingestion

Week 1: ADF Fundamentals

- ADF Overview
- Linked Services
- Datasets
- Copy Activity
- Pipeline Structure

Week 2: Dynamic Pipelines

- Parameters & Variables
- Expressions
- ForEach Activity
- Triggers
- Event-based Triggers

Week 3: Data Flows & Error Handling

- Data Flows
- Incremental Loads
- SCD in ADF
- Error Handling
- Monitoring

Week 4: Production Excellence

- Optimization
- Cost Control
- CI/CD
- Testing
- Security

PROJECT: Complete SAT Event Pipeline

CHAPTER 7: MONTH 4 - PYSPARK & DATABRICKS

Goal: Handle large-scale data | SAT: Layer 5 AI Foundation

Week 1: Spark Fundamentals

- Spark Architecture
- Databricks Setup
- DataFrames
- Reading/Writing Data

Week 2: Transformations

- Column Operations
- String/Date Functions
- Aggregations
- Joins
- Window Functions

Week 3: Performance

- Catalyst Optimizer
- Partitioning
- Caching
- Broadcast Joins
- Handling Skew

Week 4: Advanced Databricks

- Delta Lake Deep Dive
- Streaming
- Workflows
- Unity Catalog
- MLflow

PROJECT: SAT Analytics Engine

CHAPTER 8: MONTH 5 - SAT MVP PROJECT

Goal: Complete SAT Platform MVP | ALL LAYERS

"This is where everything comes together."

Week 1: Foundation

- Project Planning
- Architecture Design
- Data Model Final
- Infrastructure Setup
- Security Config

Week 2: Data Pipeline

- Event Generator
- Raw Ingestion
- Data Validation
- Bronze/Silver/Gold Layers

Week 3: Analytics

- Synapse Setup
- Analytics Queries
- Anomaly Detection
- Dashboards
- Alerts

Week 4: Polish & Launch

- Performance Testing
- Cost Analysis
- Documentation
- Demo Prep
- MVP Launch

CHAPTER 9: MONTH 6 - INTERVIEW & LAUNCH

Goal: 30 LPA Interview Ready + SAT Production

Week 1: Technical Interview

- SQL scenarios
- Data modeling
- ETL scenarios
- Spark questions
- Azure services
- Mock Interview 1

Week 2: System Design

- Scalability patterns
- Data platform design
- Real-time systems
- Governance
- Cost optimization
- Mock Interview 2

Week 3: Behavioral

- STAR method
- Project stories
- Conflict resolution
- Leadership
- Company research
- Mock Interview 3

Week 4: Final Prep

- Resume polish
- LinkedIn optimization
- Portfolio creation
- Salary negotiation (30 LPA target)
- Final mock
- SAT Production Release

CHAPTER 10: RESOURCES & REFERENCES

SQL Practice

• LeetCode SQL - leetcode.com/problemset/database/

• DataLemur - datalemur.com

• HackerRank SQL - hackerrank.com/domains/sql

Azure & Cloud

• Microsoft Learn - learn.microsoft.com/azure/

• Azure Free Account - $200 credit

• ADF Documentation

Databricks & Spark

• Databricks Community Edition - FREE

• Databricks Academy - FREE courses

• Spark: The Definitive Guide (Book)

Interview Prep

• Pramp - pramp.com (FREE mocks)

• System Design Primer (GitHub)

• Glassdoor

APPENDIX: WEEKLY TRACKER

Week	Month	Hours	Done?
Week 1	Month 1	___/14	[ ]
Week 2	Month 1	___/14	[ ]
Week 3	Month 1	___/14	[ ]
Week 4	Month 1	___/14	[ ]
Week 5	Month 2	___/14	[ ]
Week 6	Month 2	___/14	[ ]
Week 7	Month 2	___/14	[ ]
Week 8	Month 2	___/14	[ ]
Week 9	Month 3	___/14	[ ]
Week 10	Month 3	___/14	[ ]
Week 11	Month 3	___/14	[ ]
Week 12	Month 3	___/14	[ ]
Week 13	Month 4	___/14	[ ]
Week 14	Month 4	___/14	[ ]
Week 15	Month 4	___/14	[ ]
Week 16	Month 4	___/14	[ ]
Week 17	Month 5	___/14	[ ]
Week 18	Month 5	___/14	[ ]
Week 19	Month 5	___/14	[ ]
Week 20	Month 5	___/14	[ ]
Week 21	Month 6	___/14	[ ]
Week 22	Month 6	___/14	[ ]
Week 23	Month 6	___/14	[ ]
Week 24	Month 6	___/14	[ ]
Week 25	Month 7	___/14	[ ]
Week 26	Month 7	___/14	[ ]

FINAL WORDS

This is not a side project. This is not a random startup.

This is a nation-impact system design.

Great systems are built: Slowly. Quietly. Correctly.

"Dreams fail when Vision > Discipline. Your image shows structure. Now execute."

START DATE: January 20, 2025

END GOAL: 30 LPA + SAT MVP

COMMITMENT: 2 hours daily, 180 days

You're not lost - you're standing at a fork. Now take the first step.